

Chapter 14

Financial Crises, Stabilization, and Deficits

Lecturer: Mihyun You

Department of Economics
Georgetown University

Summer 2024



GEORGETOWN UNIVERSITY

Contents

- 1 The Stock Market and the Housing Market
- 2 Financial Crises
- 3 Time Lags
- 4 Government Deficit Issues



GEORGETOWN UNIVERSITY

Importance of the Stock and the Housing Market

- Introductory macroeconomics textbooks written before 1990 could largely ignore the stock and housing markets.
- Especially, financial markets were considered as a mere consequence of the *real* economy, thus a shadow of the real economy, which was believed that financial markets cannot affect the real economy.
- In the late 1990s and in the period after 2000, the fluctuations in stock and housing markets contributed to cycles in the real economy (**household wealth effect**).

Stocks and Bonds

- When a firm wants to make a large purchase to build a new factory or buy machines (investment), it often cannot pay for the purchase out of its own funds.
- Three main ways to "finance" the investment
 - ① To borrow from a bank
 - ② To issue a bond

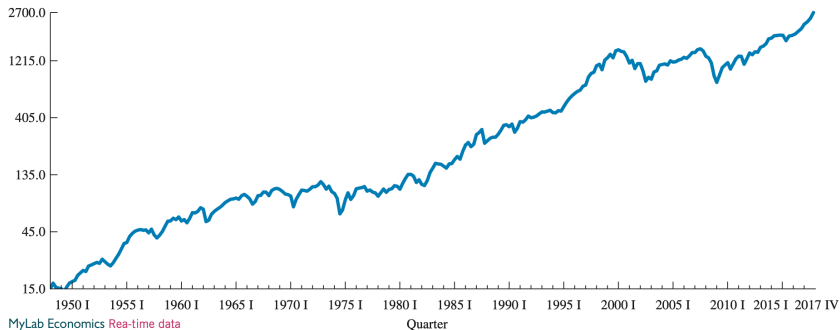
NOTE: Bond

- a promise to pay a fixed coupon periodically during the term of the bond and then to repay the full amount of the bond at the end of its term.
- ③ To issue stock

NOTE: Stock

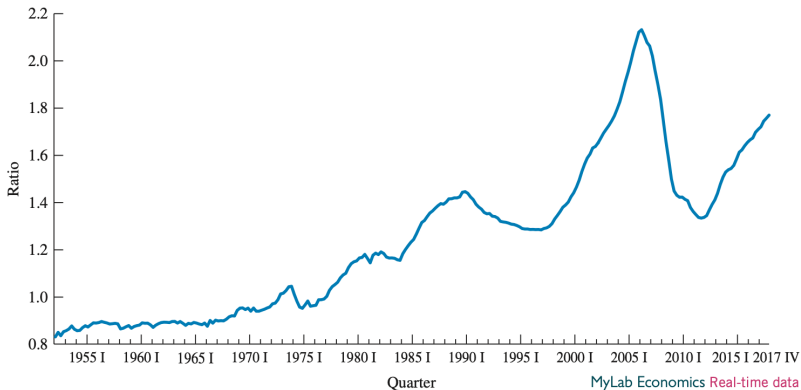
- a certificate that certifies ownership of a certain portion of a firm

The Stock Market Since 1948



▲ **FIGURE 14.1** The S&P 500 Stock Price Index, 1948 I–2017 IV
The scale is the log scale.

Housing Prices Since 1952



▲ **FIGURE 14.3** Ratio of a Housing Price Index to the GDP Deflator, 1952 I–2017 IV

Household Wealth Effects on the Economy

- Both stock and housing market have periods of large unpredictable ups and downs.
- How do these swings affect the economy?
- *Wealth* is the one of the important factors affecting consumption expenditures.
 - As a rough rule of thumb, a \$1.00 change in the value of wealth (either stocks or housing) leads to about a \$0.03 to \$0.04 change in consumer spending per year.
- An increase in stock prices may also increase investment.
- The higher the price of the firm's stock, the more money it can get per additional share issued.
- A firm can undertake more investment projects the higher the price of the stock.

From a Housing Problem to a Financial Crisis

- Falling housing prices beginning in 2006 started a process leading to the financial crisis.
- Prior to that point in time U.S. housing prices increased due to low interest rates, high demand and **mortgages lenders** willing to lend to risky borrowers through what is known as **subprime mortgages (subprimes)**.

What is a Mortgage?

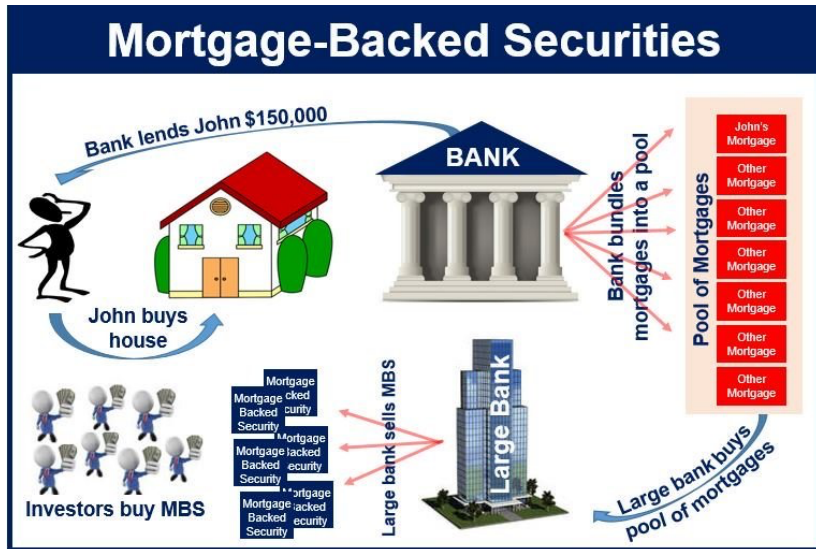


- A loan used to purchase or maintain a home, land, or other types of real estate, secured by the property itself
- The property serves as **collateral** to secure the loan.
- Mortgage applications go through a rigorous underwriting process, including credit checks and assessments of the capability of repayment of the borrower.

Terminologies

- **Securitization:** the procedure where an issuer designs a marketable financial instrument by merging or pooling various financial assets into one group.
- In theory, any financial asset can be securitized—that is, turned into a tradeable, fungible item of money value.
- **Collateralized Debt Obligation (CDO):** a complex structured financial product that is backed by a pool of loans and other assets.
- A CDO is a particular type of derivative because its value is derived from another underlying asset.
- These underlying assets become the collateral if the loan defaults.

Mortgage-Backed Securities (MBS)



Understanding the Risky Structure of Subprime Mortgages

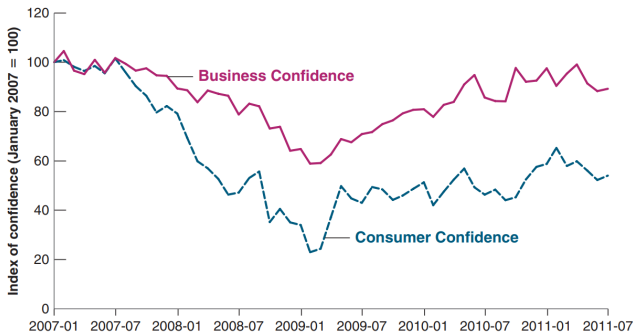
- Technically, a pool of mortgages has a more predictable repayment profile than an individual mortgage, because repayment of the latter is dependent on the individual circumstances of the mortgage borrower.
- In principle, the advent of MBS should have reduced the cost of mortgage borrowing by making mortgage financing more attractive to lenders.
- It turned out that the mortgages offered were in fact much riskier than either the lender pretended or the borrower understood.
- CDOs divided up the payments from a pool of mortgages into different streams—for example a senior tranche paid first, and junior tranches paid afterwards (as long as sufficient funds were available).

Securitization and Illiquidity of Assets

- The complexity of CDOs can increase substantially with further securitization: There are CDOs on CDOs and so on.
- Securitization created a risk that evidently was not well understood: The assessed value of the payment streams was **contingent on housing prices *continuing to rise***.
- After 2006, housing prices began to decline. Once the decline occurred many mortgages were considered to be **underwater**: this is when the value of the mortgage (debt, liability) exceeds the value of the house (asset, wealth).
- This in turn caused many borrowers to default on their mortgages thus creating a large loss for many banks.
- The securitized assets were complex and hard to value, therefore banks had to sell some assets at fire sale prices.

Macroeconomic Implications

- The immediate effects of the crisis were increasing interest rates which lowered spending and plummeting consumer confidence. The result was a sharp leftward shift in the IS curve.



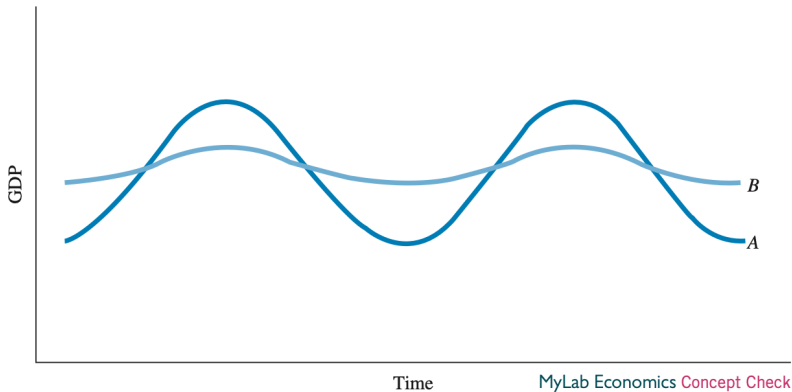
Policy Responses

IS – LM Model Analysis

- The government implemented both fiscal and monetary policy responses to the crisis.
- These responses included:
 - ① The provision of higher liquidity to the financial system.
 - ② An increase in deposit insurance limits from \$100k to \$250k to prevent bank runs.
 - ③ Passed the **Troubled Asset Relief Program (TARP)** to clean up problem assets from banks (called the bank bailout bill).
 - ④ Implemented **unconventional monetary policy** given the fed funds rate was zero.
 - ⑤ Passed the **American Recovery and Reinvestment Act** which temporarily reduced taxes to encourage consumption and increase government spending.
- The net effect of these policies was to shift the *IS* curve back to the right and shift the *LM* curve down.

Stabilization Policy

- The goal of stabilization policy is to smooth out fluctuation in GDP as much as possible.



MyLab Economics Concept Check

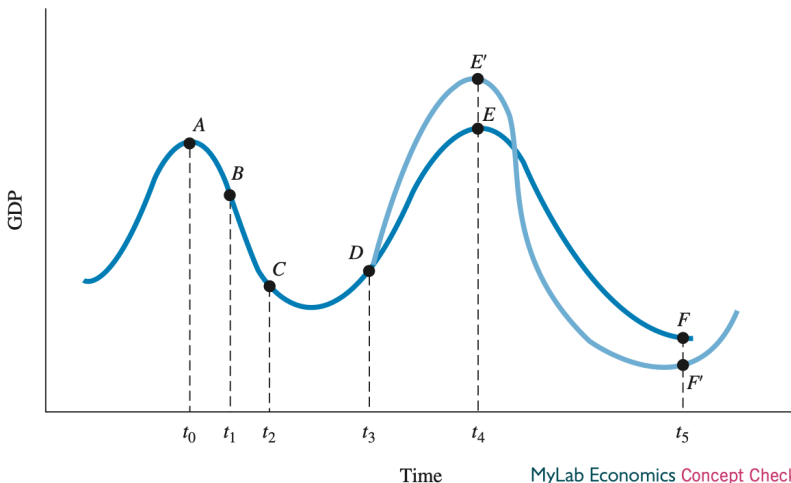
Time Lags Regarding Monetary and Fiscal Policy

- Stabilization goals are not easy to achieve, particularly in the light of various kinds of time lags.
- Economists generally recognize three kinds of time lags: recognition lags, implementation lags, and response lags.

NOTE: Three Kinds of Time Lags

- Recognition Lags: The time it takes for policymakers to recognize the existence of a boom or a slump
- Implementation Lags: The time it takes to put the desired policy into effect once economists and policymakers recognize that the economy is in a boom or a slump.
- Response Lags: The time that it takes for the economy to adjust to the new conditions after a new policy is implemented; the lag that occurs because of the operation of the economy itself.

Possible Stabilization Timing Problems



MyLab Economics Concept Check

Debates on Deficit Targeting

- Debates about deficits have been around for a long time.
 - In the 1980s the U.S. government was spending much more than it was receiving in taxes.
 - In response to the large deficits, in 1986 the U.S. Congress passed and President Reagan signed the *Graham-Rudman-Hollings Act*, referred to as GRH.
 - GRH set a target for reducing the federal deficit by a set amount each year.
- Balanced budget rules require that the government not run a deficit ever.
- Is this good policy?

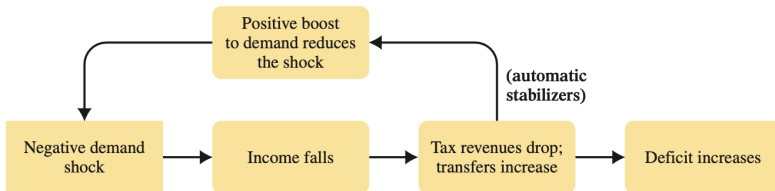
Automatic Stabilizer

- Consider a leftward shift of the AD curve caused by some negative demand shock.
- In a world without deficit targeting, the increase in the deficit during contractions provides an automatic stabilizer for the economy.
- Thus, the decrease in aggregate output caused by the negative demand shock is lessened somewhat by the growth of the deficit.
- In a world with deficit targeting, the deficit is not allowed to rise.
- The contraction in the economy will therefore be larger than it would have been without deficit targeting because the initial effect of the negative demand shock is worsened by the rise in taxes or the cut in government spending.

Deficit Targeting as an Automatic Destabilizer

a. Without Deficit Targeting

MyLab Economics Concept Check



b. With Deficit Targeting

